

IMPROVED KISSING NUMBERS IN NINETEEN AND TWENTY-ONE DIMENSIONS

ABSTRACT. We prove that the kissing numbers in 19 and 21 dimensions are at least 12276 and 30779. The best previously published lower bounds were 11948, due to Ho, and 29768, due to Cohn and Li, so the gains are 328 and 1011. Both constructions start from the odd-sign kissing configurations of Cohn and Li, delete a small block of their points, and add more new points than were deleted. In 19 dimensions we cut the 20-dimensional configuration by the hyperplane orthogonal to the indicator vector of a set T of four coordinates, project 20-dimensional sign vectors into that hyperplane, and give their tails one of two lengths. This gives 12268 points, with a proof by case analysis. A second step, verified by computer, reweights the new points and restores eight deleted points, which form one orbit of a group of 128 sign changes; the new points are then the orbits of 14 integer vectors under this group. In 21 dimensions the new points are weighted sign vectors whose sign patterns meet all eight cosets of a $[21, 9, 6]$ subcode of the punctured Golay code. They are the orbits of 192 integer vectors under a group of 16 sign changes, and the proof is an exact computation, reduced by that symmetry. We have also formalized both results in Lean 4 with Mathlib.

1. INTRODUCTION

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [3, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set X of unit vectors in $\mathbb{R}^n$ with $\langle x, y \rangle \leq 1/2$ for all distinct $x, y \in X$. We write τ_n for this maximum, as in [3, Ch. 13]. It is known only for $n \in \{1, 2, 3, 4, 8, 24\}$; see [2, Table 1.1] for references.

For $17 \leq n \leq 23$ the best constructions were Leech’s cross sections of the Leech lattice minimal vectors [5] until Cohn and Li [2] improved dimensions 17 through 21. In dimensions 19 to 21 their construction is as follows. Take the classical configuration built from a constant-weight code [6], and put an odd number of minus signs on each codeword instead of an even number. Every sign vector of a code orthogonal to the constant-weight code then has inner product at most 6 with the old points, while the kissing condition allows $\frac{1}{2}\sqrt{8n}$, which is larger for $n \geq 19$. So these sign vectors can be added, one per word of a code with minimum distance at least $n/4$. Cohn and Li obtained $\tau_{19} \geq 11692$ in this way, and Ho [4] found a larger code of 1280 words for the same purpose, reaching $\tau_{19} \geq 11948$.

Theorem 1.1. *We have $\tau_{19} \geq 12276$ and $\tau_{21} \geq 30779$.*

Table 1 compares these values with the best published lower bounds and the known upper bounds.

The mechanism is the slack between 6 and $\frac{1}{2}\sqrt{8n}$. Cohn and Li’s added points are flat: each is a sign vector scaled to squared norm 8, and none of them touches an old point. We use this slack by changing the shape of the added points.

- In 19 dimensions (Section 3) the added points come from 20 dimensions. Projecting them into a hyperplane leaves a short three-dimensional tail, and we multiply it by 9/11 or 7/5 according to its direction. This lets two added points differ in only four

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TABLE 1. Kissing number bounds in dimensions 19 and 21, as of October 2026 [1]. The previous lower bounds are due to Ho [4] in dimension 19 and to Cohn and Li [2] in dimension 21. The upper bounds are the floors of numerical semidefinite programming values [7] and are not interval-certified.

dimension	previous lower bound	new lower bound	upper bound
19	11948 [4]	12276	24417 [7]
21	29768 [2]	30779	53524 [7]

head coordinates when their tails differ, which a code of minimum distance at least 5, such as Ho's, must avoid. We add 1792 points where Ho adds 1280. The long tails come too close to 192 old points, which we delete, for a net gain of $1792 - 192 - 1280 = 320$. Eight of the 192 can be restored (Section 3.5) if the 1792 added points are given new, numerically found weights. The eight form one orbit of a group of 128 sign changes that maps the whole configuration to itself, and this group reduces the check to 14 representatives. The net gain is then $1792 - 184 - 1280 = 328$.

- In 21 dimensions (Section 4) each added point carries its own positive weight on each coordinate. That freedom allows sign patterns from the even cosets of a $[21, 9, 6]$ code as well as from the odd ones, where Cohn and Li use only the four odd cosets. The 3072 new points are the images of 192 integer vectors under a group of 16 sign changes. Half of the group is an extra sign change that fixes the octad where deletions happen, so the second half of the table needs no further deletions. The weights overshoot that octad, and we delete 13 points on it, for a net gain of $3072 - 13 - 2048 = 1011$.

The proofs differ in character. The 12268-point configuration (Proposition 3.4) has a case analysis that can be checked by hand, given the finite facts about a code of size 4096 stated in Lemma 3.1. The step to 12276 and the 21-dimensional proof depend on computer calculation in an essential way, unlike the proofs in [2]. Our added points in these two steps are integer vectors with entries up to 4616 in 19 dimensions and between $1.5 \cdot 10^8$ and $2.5 \cdot 10^8$ in absolute value in 21 dimensions. They were obtained by numerical optimization and rounding, not by hand. The theorems are statements about these explicit tables. Using their symmetry, their proofs reduce to lists of about 160 000 and 5.6 million exact integer inequalities, which we check by computer. A second program, written independently and without using the symmetry, checks all 75 343 950 and 473 658 031 pairs, and we also formalized both results in Lean 4 with Mathlib (Section 5). The formal proof does not verify our case analysis. It verifies the conclusion, that explicit sets of 12276 and 30779 unit vectors with pairwise inner products at most $1/2$ exist.

2. THE ODD-SIGN CONFIGURATIONS

We identify a binary word with its support. For $c \in \mathbb{F}_2^n$ we write $\text{wt}(c)$ for its weight, $(-1)^c \in \{\pm 1\}^n$ for its sign vector, and $\lambda \circ q$ for the coordinatewise product of two vectors.

Let $\mathcal{O}$ be a set of words of length n with weight 8 and pairwise intersections of size at most 4. The *odd configuration* $B(\mathcal{O})$ consists of the $4\binom{n}{2}$ roots $\pm 2e_i \pm 2e_j$ and, for each $o \in \mathcal{O}$, the 128 vectors with entries ± 1 on o , zeros elsewhere and an odd number of minus signs [2, §2]. All have squared norm 8. We call these 128 vectors the *octad vectors* on o , and their set the *fiber* over o .

Lemma 2.1 (Cohn and Li). *Distinct $b, b' \in B(\mathcal{O})$ satisfy $\langle b, b' \rangle \leq 4$.*

Proof. Two distinct roots have inner product in $\{0, \pm 4, -8\}$. A root has two nonzero coordinates, each contributing at most 2 against an octad vector, so that inner product is at most 4. Two vectors on the same octad have odd sign patterns, so they differ in an even and positive number $d \geq 2$ of places and have inner product $8 - 2d \leq 4$. Two vectors on different octads share at most four coordinates. $\square$

The next lemma is the reason for odd signs. It is the weighted form of the argument in [2, §2].

Lemma 2.2. *Let $c \in \mathbb{F}_2^n$ meet every word of $\mathcal{O}$ in an even number of coordinates, let $q = (-1)^c$, and let $\lambda \in \mathbb{R}_{>0}^n$. Let b be an octad vector on $o \in \mathcal{O}$. Then q and b disagree in an odd number of coordinates of o , and*

$$\langle \lambda \circ q, b \rangle \leq \sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j,$$

with equality for some b on o . In particular, for $\lambda = \mathbf{1}$ we have $\langle q, b \rangle = 6$ if q and b disagree in exactly one coordinate and $\langle q, b \rangle \leq 2$ otherwise. For a root $b = \pm 2e_i \pm 2e_j$ the inner product is at most $2(\lambda_i + \lambda_j)$.

Proof. The sign vector q has an even number of minus signs on o , and b has an odd number. So they disagree in an odd number of coordinates of o , in particular in at least one, and the bound follows. Equality holds for the odd pattern that agrees with q except at one coordinate where λ is smallest. For $\lambda = \mathbf{1}$, d disagreements give $\langle q, b \rangle = 8 - 2d$ with d odd. $\square$

Throughout we test compatibility in integers. For nonzero $x, y \in \mathbb{Z}^n$ the angle between them is at least 60° if and only if

$$(1) \quad \langle x, y \rangle \leq 0 \quad \text{or} \quad 4\langle x, y \rangle^2 \leq |x|^2 |y|^2.$$

We call such a pair *compatible*; the same test applies to rational vectors after scaling. If every pair from a finite set of nonzero rays is compatible, then the rays normalized to unit length form a kissing configuration, and distinct rays give distinct points because (1) excludes cosine 1. With $\lambda = \mathbf{1}$, Lemma 2.2 shows that the sign vector of a word that meets every word of $\mathcal{O}$ evenly is compatible with $B(\mathcal{O})$ when $6 \leq \frac{1}{2}\sqrt{8n}$, that is, for $n \geq 18$, with contacts when $n = 18$.

3. DIMENSION 19

3.1. The codes. Coordinates are $1, \dots, 20$. Let $E \subseteq \mathbb{F}_2^{20}$ be the span of the twelve words

$$\begin{aligned} &\{1, 8, 9, 12, 16, 17, 18, 19\}, \{2, 10, 11, 14, 15, 17, 18, 20\}, \{3, 7, 9, 13, 15, 16, 17, 19\}, \\ &\{4, 7, 8, 10, 12, 15, 16, 19\}, \{5, 10, 12, 13, 15, 16, 17, 18\}, \{6, 7, 8, 9, 10, 13, 16, 18\}, \\ &\{1, 4, 7, 9\}, \{1, 5, 6, 18, 20\}, \{1, 3, 12, 15, 20\}, \{1, 10, 13, 19, 20\}, \\ &\{1, 3, 5, 6, 7, 13, 14, 15, 18, 20\}, \{2, 4, 6, 7, 8, 13, 14, 16, 17, 18\}. \end{aligned}$$

On the first 19 coordinates these are Ho's generators $m_1, \dots, m_6, s_1, \dots, s_4, r_1, r_2$ [4, §2]. Our coordinate 20 is Ho's coordinate 24, the last column of his extension matrix P , so we have added it to the words where that column has a 1. Ho shows that the full extension to length 24 is the extended binary Golay code [4, Prop. 2]. So E is the Golay code punctured at Ho's coordinates 20 to 23, and its dual $E^\perp$ is the Golay code shortened at the same four coordinates [8]. The 130 words of weight 8 in $E^\perp$ are the *octads*. They form the shortened Steiner system $S(5, 8, 24)$ with $n = 20$ of [2, Table 2.1], so $B(\text{octads})$ is Cohn and Li's 17400-point configuration C_{20} in $\mathbb{R}^{20}$.

The five octads of the Golay code that contain four given coordinates partition the other twenty coordinates into five tetrads, and these six tetrads form a *sextet* [3, Ch. 10]. Puncturing at the four given coordinates turns those five octads into five words of weight 4 in E . We call these five words the *tetrads*.

The following facts are finite computations on the 4096 words of E .

Lemma 3.1. (1) *The weight enumerator of E is*

$$1 + 5z^4 + 64z^5 + 240z^6 + 320z^7 + 250z^8 + 640z^9 + 1056z^{10} \\ + 640z^{11} + 250z^{12} + 320z^{13} + 240z^{14} + 64z^{15} + 5z^{16} + z^{20}.$$

The five words of weight 4 are pairwise disjoint and cover all 20 coordinates.

- (2) *Let $T = \{1, 4, 7, 9\}$, one of the five, and let H be the other 16 coordinates, the head. For $e \in E$ write e_T and e_H for its parts on T and H . The nonzero words with $\text{wt}(e) \leq 7$ have $(\text{wt}(e_T), \text{wt}(e_H))$ equal to $(4, 0)$ for 1 word, $(0, 4)$ for 4, $(1, 4)$ for 64, $(0, 6)$ for 96, $(2, 4)$ for 144, $(1, 6)$ for 256 and $(3, 4)$ for 64.*
- (3) *Two distinct octads meet in at most 4 coordinates. The octads meet T in 0, 2 or 4 coordinates, for 30, 96 and 4 octads respectively.*

The intersections in (3) are even because $T \in E$ and the octads lie in $E^\perp$.

Corollary 3.2. *Let $c, c' \in E$ be distinct, and let $d_T = \text{wt}(c_T + c'_T)$ and $d_H = \text{wt}(c_H + c'_H)$. If $d_T = 0$, then $d_H = 4$ (and $c + c'$ is a tetrad inside H) or $d_H \geq 6$. If $1 \leq d_T \leq 3$, then $d_H \geq 4$. If $d_T = 4$, then $d_H = 0$ (and $c' = c + \mathbf{1}_T$), $d_H = 4$ or $d_H \geq 6$.*

Proof. The first two cases follow from Lemma 3.1(2) applied to $c + c'$. The case $d_T = 4$ is the case $d_T = 0$ applied to c and $c' + \mathbf{1}_T \in E$. $\square$

3.2. The configuration. Let

$$R = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

acting on the coordinates $(1, 4, 7, 9)$ of T . The rows of $R/2$ are an orthonormal basis of $\mathbf{1}_T^\perp \cap \mathbb{R}^T$, so $(R/2)^\top(R/2) = I - J/4$. The map $\varphi(v) = (v_H, Rv_T/2) \in \mathbb{R}^{16} \times \mathbb{R}^3$ therefore satisfies

$$(2) \quad \langle \varphi(u), \varphi(v) \rangle = \langle u, v \rangle - \frac{1}{4} \left(\sum_{j \in T} u_j \right) \left(\sum_{j \in T} v_j \right),$$

and it is an isometry from the hyperplane $\mathbf{1}_T^\perp$ onto $\mathbb{R}^{19}$. The configuration in $\mathbb{R}^{19}$ has two parts.

Base. There are 10668 points of C_{20} orthogonal to $\mathbf{1}_T$, as many as in Leech's 19-dimensional configuration. We delete from them the 192 points that lie on the four octads containing T ; these are the points with two plus and two minus signs on T . The remaining 10476 points b enter as $\varphi(b)$. They are the 480 roots inside H , the 12 roots $\pm 2(e_i - e_j)$ with $i, j \in T$, the 3840 odd vectors on the 30 octads disjoint from T , and the 6144 odd vectors on the 96 octads meeting T twice whose two T -entries have opposite signs.

Additions. Let $c \in E$ with $\text{wt}(c_T) \in \{1, 2, 3\}$ and $q = (-1)^c$. The projected tail $t_0 = Rq_T/2$ satisfies $|t_0|^2 = 4 - \frac{1}{4}(\sum_T q_j)^2$ by (2). When $\text{wt}(c_T)$ is odd, $t_0 \in \{\pm 1\}^3$ and $|t_0|^2 = 3$; we call c *diagonal*. When $\text{wt}(c_T) = 2$, $t_0 = \pm 2f_k$, where f_1, f_2, f_3 is the standard basis of the tail $\mathbb{R}^3$,

TABLE 2. The blocks of the 19-dimensional configuration.

block	size	squared norm
roots in H	480	8
roots in T	12	8
octad vectors, octad disjoint from T	3840	8
octad vectors, octad meeting T twice	6144	8
diagonal additions	1024	2179/121
axis additions	768	596/25
total	12268	

and $|t_0|^2 = 4$; we call c an *axis* word. The added point is

$$a(c) = (q_H, \alpha t_0), \quad \alpha = \begin{cases} 9/11, & c \text{ diagonal,} \\ 7/5, & c \text{ axis,} \end{cases}$$

with $|a(c)|^2 = 2179/121$ or $596/25$. Multiplying by 22 or 10 makes it an integer vector, and the base points $\varphi(b)$ become integer vectors after multiplying by 2.

Choice of words. Let W be the span of the five tetrads, a code of size 32. The 3584 words with $\text{wt}(c_T) \in \{1, 2, 3\}$ fall into 112 cosets of W . Each coset $c + W$ is a 5-cube: its vertices are the 32 words, and each of the five directions adds one tetrad. We keep one of the two bipartition classes of each cube, 16 words. Lemma 3.3 below holds for either class, so any choice gives a valid configuration of the same size. For definiteness (this is the choice used in the accompanying data), we keep the words c for which $c + c_{\min}$ is a sum of an even number of tetrads, where $c_{\min}$ is the smallest word of the coset read as the integer $\sum_{j \in c} 2^{j-1}$. This gives 1024 diagonal and 768 axis words, 1792 in all.

The total is $10668 - 192 + 1792 = 12268$. Table 2 lists the blocks.

3.3. Proof. Some restriction inside each cube is needed, because two words of one cube that differ by a tetrad inside H have the same T -pattern and $d_H = 4$. For two diagonal words this gives inner product $8 + 3(9/11)^2 \approx 10.01$ against the bound $\frac{1}{2}|a|^2 \approx 9.00$.

Lemma 3.3. *Two distinct kept words c, c' satisfy $c_H \neq c'_H$, and $d_H \geq 6$ if $c_T = c'_T$.*

Proof. If $c_H = c'_H$, then $c + c'$ is a word supported in T , so it is $\mathbf{1}_T$ by Lemma 3.1(2). Then c and c' lie in the same coset of W and differ by one tetrad, so they lie in opposite classes of the 5-cube and are not both kept. If $c_T = c'_T$ and $d_H = 4$, then $c + c'$ is a tetrad in H , and the same argument applies. $\square$

Proposition 3.4. *The 12268 vectors of Table 2 are pairwise compatible and span $\mathbb{R}^{19}$. Hence $\tau_{19} \geq 12268$.*

Proof. We check every pair, using the unnormalized vectors.

Base with base. By (2) the map φ preserves inner products on $\mathbf{1}_T^\perp$, so Lemma 2.1 gives $\langle \varphi(b), \varphi(b') \rangle \leq 4 = \frac{1}{2} \cdot 8$.

Base with additions. Since $\sum_T b_j = 0$, equation (2) gives

$$(3) \quad \langle a(c), \varphi(b) \rangle = \langle q_H, b_H \rangle + \alpha \langle q_T, b_T \rangle = \langle q, b \rangle + (\alpha - 1) \langle q_T, b_T \rangle.$$

The word c lies in E , so it meets every octad evenly, and Lemma 2.2 applies: for an octad vector b , $\langle q, b \rangle = 6$ if q and b disagree in exactly one place of the octad and $\langle q, b \rangle \leq 2$

TABLE 3. Largest inner product between a base point and an addition in dimension 19.

base points	count	reason	diagonal	axis
roots in H	480	$b_T = 0$	4	4
roots in T	12	$\alpha \langle q_T, b_T \rangle \leq 4\alpha$	36/11	28/5
octads disjoint from T	3840	$b_T = 0$, Lemma 2.2	6	6
octads meeting T twice	6144	three cases above	6	34/5
bound $\sqrt{2 a ^2}$			6.0014	6.9051

TABLE 4. Largest inner products between two additions in dimension 19, for the cases with nonnegative tail product. The bound is $\frac{1}{2}|a||a'|$. Cosines are rounded to four places.

types	case	$\langle a, a' \rangle$	bound	cosine
diagonal, diagonal	$d_T = 0, d_H = 6$	727/121	2179/242	0.3337
	$d_T = 2, \langle t_0, t'_0 \rangle = 1, d_H = 4$	1049/121	2179/242	0.4814
axis, axis	$d_T = 0, d_H = 6$	296/25	298/25	0.4966
	$d_T = 2, d_H = 4$	8	298/25	0.3356
diagonal, axis	$d_T = 1, d_H = 4$	566/55	$\sqrt{324671/3025}$	0.4967

otherwise. The bound to meet is $\frac{1}{2}|a|\sqrt{8} = \sqrt{2|a|^2}$, which is $\sqrt{4358/121} \approx 6.0014$ for diagonal and $\sqrt{1192/25} \approx 6.9051$ for axis additions.

For the 6144 points on octads meeting T twice, $b_T = \pm(e_i - e_j)$ with i, j the two coordinates of the octad in T , so $\langle q_T, b_T \rangle \in \{-2, 0, 2\}$. If it is 0, the inner product is $\langle q, b \rangle \leq 6$. If it is 2, the inner product is at most $6 + 2(\alpha - 1)$, which is $62/11$ or $34/5$. If it is -2 , then q and b disagree at both coordinates of the octad in T . The number of disagreements is odd, so it is at least 3, $\langle q, b \rangle \leq 2$, and the inner product is at most $2 - 2(\alpha - 1)$. Table 3 collects the maxima.

All maxima are below the bounds, since $36 \cdot 121 = 4356 < 4358$ and $(34/5)^2 = 1156/25 < 1192/25$.

Additions with additions. For kept words $c \neq c'$,

$$\langle a(c), a(c') \rangle = 16 - 2d_H + \alpha\alpha' \langle t_0, t'_0 \rangle, \quad \langle t_0, t'_0 \rangle = \langle q_T, q'_T \rangle - \frac{1}{4} \left(\sum_T q_j \right) \left(\sum_T q'_j \right).$$

Here $d_H \geq 4$ in every case, and $d_H \geq 6$ when $c_T = c'_T$, by Corollary 3.2 and Lemma 3.3. The tail product depends only on the two patterns on T :

- two diagonal words have $d_T \in \{0, 2, 4\}$ and $\langle t_0, t'_0 \rangle = 3, \pm 1, -3$ respectively;
- two axis words have $d_T \in \{0, 2, 4\}$ and $\langle t_0, t'_0 \rangle = 4, 0, -4$ respectively;
- a diagonal and an axis word have $d_T \in \{1, 3\}$ and $\langle t_0, t'_0 \rangle = 2$ or -2 respectively.

The cases with $\langle t_0, t'_0 \rangle < 0$ give at most 8, below every bound. Table 4 lists the others at the smallest allowed d_H .

Every entry is strictly below its bound. For the diagonal–axis row we compare $(566/55)^2 = 320356/3025$ with $324671/3025$. So all pairs are compatible, and the roots inside H and inside T already span $\mathbb{R}^{19}$. $\square$

3.4. Remarks. The tail factors are nearly forced. A diagonal addition against an octad vector on an octad disjoint from T needs $36 \leq 2(16 + 3\alpha^2)$, so $\alpha \geq \sqrt{2/3} \approx 0.8165$ for diagonal words. Two axis additions with the same pattern at $d_H = 6$ need $4 + 4\alpha^2 \leq \frac{1}{2}(16 + 4\alpha^2)$, so $\alpha \leq \sqrt{2} \approx 1.4142$ for axis words. The diagonal-axis pairs at $d_T = 1$, $d_H = 4$ couple the two factors. Numerically, the conditions behind Tables 3 and 4 allow the diagonal factor in about $[0.8165, 0.877]$ and the axis factor in about $[1.304, 1.4142]$, in a thin region where a larger axis factor permits a larger diagonal factor. We chose $9/11$ and $7/5$ as simple fractions inside it. With plain projection, both factors equal to 1, the diagonal-axis pairs have inner product 10 against $\frac{1}{2}\sqrt{19 \cdot 20} \approx 9.75$, so plain projection fails.

Axis factors above 1 are what the 192 deleted points cannot tolerate. A deleted point has $b_T \in \{\pm 1\}^4$ with two signs of each kind. An axis word with $q_T = b_T$ whose sign vector disagrees with b in exactly one place of the octad gives inner product $6 + 4 \cdot \frac{2}{5} = 38/5 > 6.9051$. A computer check shows that each deleted point conflicts in this way with some kept axis addition.

We have not determined the largest configuration of this shape. One numerical observation bears on it. The 3584 candidate words have exactly 1792 distinct heads, so both bipartition classes use the same heads. A numerical search over all 2^{16} sign vectors on H , looking for a tail compatible with this 10476-point base by local optimization from 48 starting points, found one only for these 1792 heads. A local search can miss a feasible tail, so this is not a proof.

No addition has a contact. By direct computation, all 2386584 contacts (pairs with cosine exactly $1/2$) are between base points. The closest pairs involving an addition have cosine $\sqrt{1089/4358} \approx 0.49989$, a diagonal addition against an octad vector on an octad disjoint from T .

3.5. Restoring eight deleted points. Eight of the 192 deleted points can be restored if the additions are given new weights. The resulting 12276-point configuration was found by an AI agent, starting from the configuration of Proposition 3.4, and was given in other coordinates. A coordinate permutation and sign change carries its base onto ours, so we state it in the coordinates of this section. The new points keep the sign patterns of the 1792 additions.

Base. Let $u = \{3, 6, 8, 19\}$, so that $T \cup u$ is one of the four octads containing T . Let P be the set of the eight vectors

$$p_y = y + e_1 + e_4 - e_7 - e_9,$$

where y has entries ± 1 on u , zeros elsewhere and an odd number of minus signs. Each p_y is an octad vector on $T \cup u$ with two plus and two minus signs on T , so P is contained in the 192 deleted points. Their images are $\varphi(p_y) = (y_H, 2f_1)$. Two of them have inner product 0 if they differ in all four places of u and 4 otherwise. The base consists of the 10476 points of Table 2 and the eight points $\varphi(p_y)$, 10484 points in all.

Sign group. For a word $g \in E$ that misses T , let S_g change the signs of the coordinates in g . It preserves $\mathbf{1}_T^\perp$, and $\varphi \circ S_g = S'_g \circ \varphi$, where S'_g changes the signs of the head coordinates in g ; we write S_g for both. Let $\Gamma \subseteq E$ be the span of the seven words

$$\begin{aligned} \{2, 5, 10, 11, 13, 15\}, \quad \{2, 5, 10, 16, 17, 20\}, \quad \{2, 5, 12, 14, 15, 17\}, \quad \{2, 10, 13, 14, 16, 18\}, \\ \{2, 3, 6, 10, 14, 15\}, \quad \{2, 3, 8, 10, 11, 17\}, \quad \{2, 3, 11, 13, 16, 19\}. \end{aligned}$$

It has 128 words: 0, 48 of weight 6, 30 of weight 8, 48 of weight 10, and H . Every word of Γ misses T . The words of E that miss T form a group of order 256, which contains the four tetrads other than T , and Γ is one of the 16 subgroups of index 2 in it that contain none of these tetrads. The first four generators miss u ; the last three meet u in $\{3, 6\}$, $\{3, 8\}$ and $\{3, 19\}$. The sign changes S_g , $g \in \Gamma$, form a group $\{S_g\}$ of order 128.

Lemma 3.5. *Every S_g , $g \in \Gamma$, maps the 10484-point base onto itself. On the 48 deleted points of each fiber over an octad containing T , the group $\{S_g\}$ has six orbits of size 8: the sets of points of that fiber with a given sign pattern on T . In particular P is one orbit.*

Proof. A word $g \in E$ meets every octad evenly, so S_g maps roots to roots and each fiber onto itself. Since g misses T , S_g preserves $\mathbf{1}_T^\perp$ and fixes the entries on T . So S_g maps the points of each fiber that are orthogonal to $\mathbf{1}_T$ onto themselves, and it maps the 10476 base points of Proposition 3.4 onto themselves. Let $T \cup u'$ be an octad containing T . The deleted points on it with a given pattern on T are the eight sign vectors on u' with a fixed parity of minus signs, and S_g acts on them through $g \cap u'$, which has even size. For each of the four tetrads u' , the map $g \mapsto g \cap u'$ takes all eight even subsets of u' as values on Γ ; this is a finite computation. So $\{S_g\}$ acts transitively on these eight points. The set P is the class of the pattern $(+, +, -, -)$ on $T \cup u$, so it is an orbit, and the base is mapped onto itself. $\square$

New points. Let $r_1, \dots, r_{14} \in \mathbb{Z}^{19}$ be the vectors of Table 5 in Appendix A. Their entries have absolute values at most 4616, the only zero entries are in the tail, and their squared norms lie between 63 978 448 and 64 002 793. The new points are the $14 \cdot 128 = 1792$ vectors $S_g(r_i)$, $g \in \Gamma$.

They are reweighted additions. For each new point x there is a kept word c such that x has the sign of $q = (-1)^c$ on every head coordinate and, among the 14 tail directions $\{\pm 1\}^3 \cup \{\pm 2f_k\}$, its tail is closest in angle to $t_0 = Rq_T/2$. Every kept word occurs exactly once. So $r_1, \dots, r_6$ are axis words and $r_7, \dots, r_{14}$ are diagonal words, and Γ permutes the kept words. Define the tail factor of a representative as the length of its tail divided by $|t_0|$ and by the root mean square of its head entries. It is between 0.833 and 0.872 for diagonal and between 1.234 and 1.414 for axis words, where Section 3 uses 9/11 and 7/5. Remark 3.9 says where the weights differ from those of Section 3.

We do not have a closed form for these weights. They were found by numerical optimization with the symmetry under Γ imposed: a penalty on pairs with cosine above 1/2 was minimized over all 19 coordinates of the 14 representatives, 266 real variables, and the result was rounded to integers at scale 8000. The proof does not depend on how they were found.

The total is $10476 + 8 + 1792 = 12276$, that is, $10668 - 184 + 1792$.

Proposition 3.6. *The 10484 base points and the 1792 vectors $S_g(r_i)$ are 12276 pairwise compatible nonzero vectors spanning $\mathbb{R}^{19}$. Hence $\tau_{19} \geq 12276$.*

Each S_g is orthogonal and $S_g S_{g'} = S_{g+g'}$. With Lemma 3.5, every pair involving a new point is carried by some S_g to a pair in one of these finite lists:

- (i) the $\binom{14}{2} \cdot 128 + 14 \cdot 127 = 13\,426$ products $\langle r_i, S_g r_j \rangle$ with $i \leq j$, excluding $i = j$, $g = 0$;
- (ii) the $14 \cdot 10484 = 146\,776$ products $\langle r_i, b \rangle$ with b in the base.

Lemma 3.7. *Let $L = 50\,000$. For every product $\langle x, y \rangle$ in lists (i) and (ii), $\langle x, y \rangle \leq (\frac{1}{2} - \frac{1}{L})|x||y|$. In particular every such pair satisfies (1).*

Proof. This is a finite computation in exact integer arithmetic. For $\langle x, y \rangle > 0$ the inequality is equivalent to $(2L\langle x, y \rangle)^2 \leq (L - 2)^2|x|^2|y|^2$, and we verified this for all 160 202 products. As an independent check, a second program, written separately and without the symmetry reduction, verified (1) for all 75 343 950 pairs of the configuration, and so does the formal proof of Section 5. Both programs are available with the data (see Data availability). $\square$

Proof of Proposition 3.6. The base consists of points of C_{20} orthogonal to $\mathbf{1}_T$, so base pairs are compatible by Lemma 2.1 and (2). Pairs involving a new point are compatible by Lemma 3.5 and Lemma 3.7. The base contains all 492 roots, which span $\mathbb{R}^{19}$. $\square$

This proves Theorem 1.1 for $n = 19$.

3.6. Remarks on the restoration.

Remark 3.8. In the 12268-point configuration the largest cosine over pairs involving an addition is 0.49989. In the 12276-point configuration the smallest value of $\frac{1}{4} - \cos^2$ over such pairs is $5393/170666864 \approx 3.16 \cdot 10^{-5}$ for base–new pairs and about $2.17 \cdot 10^{-5}$ for new–new pairs, where the largest cosine is $31\,997\,485/63\,997\,753 \approx 0.4999783$, between r_3 and one of its images. No pair involving a new point has cosine within 10^{-5} of $1/2$; 1792 base–new and 3456 new–new pairs are within 10^{-4} . All 2 389 952 contacts are between base points, and 3368 of them involve a point of P .

Remark 3.9. In the 12268-point configuration each deleted point conflicts with the axis additions whose sign vector agrees with it on T and disagrees with it in exactly one place of its octad (inner product $38/5$). For the points of P these are all 128 axis additions with tail along $+f_1$, and each of them conflicts with four points of P ; no other addition conflicts with P . These 128 points form the orbit of r_6 . So restoring all of P asks for the same change in the table as restoring one point of it. In the table, r_6 has tail factor 1.235, and its entries on 3, 6, 8, 19 are 1453 in absolute value against 1777 on the other head coordinates, a ratio of 0.82. The other axis representatives have tail factor 1.400 (tail along $-f_1$) or 1.414 (tail along $\pm f_2$, $\pm f_3$), and the diagonal ones 0.834 or 0.872 as the first tail coordinate is negative or positive. Eight of the 14 representatives have all 16 head entries equal in absolute value, and the head entries of each of the others except r_6 vary by at most 1.6%. So the table differs from the additions of Section 3 mainly in its tail factors and in the orbit of r_6 .

Remark 3.10. For each of the 184 points still deleted, the largest cosine with a new point lies between 0.5286 and 0.5525. By Lemma 3.5, a point restored on any fiber forces its whole orbit of 8. The same agent ran numerical searches for a second orbit of 8 under Γ , for orbits of 4 or 8 points under subgroups of order 64, for a whole fiber of 48 points, and for one more point under a subgroup of order 16. None found a valid table. We do not know whether one exists.

Remark 3.11. Among groups of sign changes S_g with $g \in E$ missing T , the group Γ is as large as possible for tables like ours. These words g form a group of order 256, and it contains the four tetrads u' other than T . If $S_{u'}$ were in the group, a new point x and $S_{u'}x$ would both be in the configuration, with cosine $1 - 2s/|x|^2$, where s is the part of $|x|^2$ on u' . For every r_i and every such u' , $s < 0.23|r_i|^2$, so this cosine is above 0.54. So a group that avoids the four tetrads has order at most 128, and with it the table needs at least 14 representatives.

The numbers in Remarks 3.8–3.11 were computed by computer; the fractions are exact and the decimals are rounded.

4. DIMENSION 21

4.1. The codes. Coordinates are $1, \dots, 24$, and the construction lives on the first 21. Let $\mathcal{G}$ be the extended binary Golay code: the cyclic code of length 23 generated by $1 + t + t^5 + t^6 + t^7 + t^9 + t^{11}$, with the coefficient of t^k at coordinate $k + 1$, extended by an overall parity bit at coordinate 24 [3, Ch. 3, §2.8]. Let $\mathcal{O}$ be the 210 octads of $\mathcal{G}$ that vanish on coordinates 22, 23 and 24. Then $B = B(\mathcal{O})$ is Cohn and Li's 27720-point configuration in $\mathbb{R}^{21}$.

Let D be $\mathcal{G}$ punctured to the first 21 coordinates, a $[21, 12, 5]$ code. Because $\mathcal{G}$ is self-dual and the octads in $\mathcal{O}$ vanish on the punctured coordinates, every word of D meets every octad of $\mathcal{O}$ evenly. We call the 21 words of weight 5 in D *lines*. Two sign vectors of words of D at distance d have cosine $1 - 2d/21$, which exceeds $1/2$ only for $d = 5$, where it is $11/21$. Cohn

and Li use the 2048 odd-weight words of D . The sum of two odd words is even, so no two of them are at distance 5.

Let $C \subseteq D$ be the span of the nine words (bit j of the binary expansion is coordinate $j + 1$)

1559917, 15163, 774651, 1540761, 485602, 408336, 678478, 1748752, 1231514.

It is a $[21, 9, 6]$ code with only even weights, and D/C has eight cosets. Four are even, with representatives 0, 1393, 2139, 3370, and contain no lines. Four are odd, with representatives 696, 1993, 2787, 3986, and contain 6, 3, 6 and 6 lines. These are finite computations on the 4096 words of D . The code C comes from an earlier, unpublished 30280-point configuration found by an AI agent.

The following lemma is not used in the proof. It explains why the new points must be weighted.

Lemma 4.1. *Let c, c' be distinct words of D . If they lie in the same coset of C , or in two cosets of the same parity, then $\text{wt}(c + c') \geq 6$. If $c + C$ and $c' + C$ have opposite parity, then their sum is an odd coset, which contains a line; so $\text{wt}(c_1 + c'_1) = 5$ for some $c_1 \in c + C$ and $c'_1 \in c' + C$.*

Proof. In the first case $c + c'$ lies in C or in an even coset, and the even words of D have weight at least 6. In the second case $c + c'$ lies in an odd coset, and every odd coset contains a line. $\square$

So flat sign vectors from an even coset and an odd coset conflict at cosine $11/21$.

4.2. Weighted sign vectors. The simplest way to avoid this conflict is to weight a line. Let ℓ be a line, and put $\lambda_j = \sqrt{2/5}$ for $j \in \ell$ and $\lambda_j = 1$ otherwise, so $|\lambda|^2 = 18$. For $c \in D$ the point $\lambda \circ (-1)^c$ has, by Lemma 2.2, inner product at most $\sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j$ with the octad vectors on o . An octad meets ℓ in 0, 2 or 4 coordinates, because $\ell \in D$, and the bound is then $8 - 2 = 6$, $6 + 2\sqrt{2/5} - 2\sqrt{2/5} = 6$, or $4 + 2\sqrt{2/5} < 6$. This is at most $\frac{1}{2}\sqrt{8}|\lambda| = 6$. So among weight vectors equal to 1 off a line and constant on it, $\sqrt{2/5}$ is the smallest weight on the line that the base allows. A small weight on ℓ weakens the conflicts at distance 5 that run through ℓ .

In general, let $c \in D$, $q = (-1)^c$ and $\lambda \in \mathbb{R}_{>0}^{21}$. Lemma 2.2 shows that $\lambda \circ q$ is compatible with every point of B exactly when

$$(4) \quad \sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j \leq \sqrt{2}|\lambda| \quad \text{for all } o \in \mathcal{O}, \quad \lambda_i + \lambda_j \leq \frac{1}{\sqrt{2}}|\lambda| \quad \text{for all } i \neq j.$$

The hypothesis $c \in D$ is what makes the first condition exact. Two points $\lambda \circ q$ and $\mu \circ q'$ have inner product $\sum_j \lambda_j \mu_j (-1)^{(c+c')_j}$, which depends only on the weights and on $c + c'$.

4.3. The configuration. *Base.* Let $Q = \{3, 6, 10, 11, 16, 18, 20, 21\} \in \mathcal{O}$. Delete from B the 13 odd sign patterns on Q whose signs, in the coordinate order of Q , are

$$\begin{aligned} &+---+---+, +---+---+, +-+---+---, +-+---+---, +-+---+---, \\ &+-+---+---, +-+---+---, +-+---+---, +-+---+---, +-+---+---, \\ &+++++---, +++++---, +++++---, +++++---, +++++---. \end{aligned}$$

We know no closed rule for them; they are exactly the odd patterns on Q that have cosine above $1/2$ with some new point (Section 4.5). The rest, B' , has $840 + 209 \cdot 128 + 115 = 27707$ points.

Sign group. Let $\Sigma \subseteq D$ be the span of 92361, 20689, 346115 and $w = 14723$, with supports

$$\{1, 4, 7, 8, 12, 14, 15, 17\}, \quad \{1, 5, 7, 8, 13, 15\}, \quad \{1, 2, 12, 15, 17, 19\}, \quad \{1, 2, 8, 9, 12, 13, 14\}.$$

For $h \in \Sigma$ let S_h change the signs of the coordinates in h . The group $\{S_h\}$ has order 16. The first three generators span a subgroup $\Sigma_0 \subseteq C$ of order 8; the word w has odd weight, so $w \notin C$, but $w \in D$. Section 4.5 explains where w comes from. All four generators miss Q , so every S_h fixes every vector supported on Q , including the 13 deleted points.

New points. Let $r_1, \dots, r_{192} \in \mathbb{Z}^{21}$ be the vectors listed in the accompanying data (see Data availability). They have no zero entries, their entries have absolute values between $149\,071\,270$ and $249\,507\,763$, and their squared norms lie between $0.999999998 \cdot 10^{18}$ and $1.000000002 \cdot 10^{18}$. Appendix A gives one of them for each coset below. The new points are the $192 \cdot 16 = 3072$ vectors $S_h(r_i)$.

The sign words of $r_1, \dots, r_{192}$ and their Σ_0 -images (the words c with $\text{sign}(r) = (-1)^c$), 1536 words in all, fill half of each of the six cosets $c_0 + C$ with

$$c_0 \in \{696, 1993, 2787\} \text{ (odd)}, \quad c_0 \in \{1393, 2139, 3370\} \text{ (even)},$$

with 32 representatives and 256 words on each. Adding w swaps the cosets in the pairs $0 \leftrightarrow 696$, $1393 \leftrightarrow 1993$, $2139 \leftrightarrow 2787$ and $3370 \leftrightarrow 3986$. So the 3072 sign words of all new points meet every coset of C : 512 words on each of 1393, 1993, 2139, 2787 and 256 on each of 0, 696, 3370, 3986. Cohn and Li's four odd cosets are all present, together with all four even ones.

On the cosets of 1993 and 2139 the weights of r_i follow those of Section 4.2, with the lines $\{8, 15, 16, 18, 19\}$ and $\{1, 3, 5, 17, 18\}$: after scaling each representative to $|\lambda|^2 = 18$, every entry is within 10^{-6} of $\sqrt{2/5}$ on the line and of 1 off it. On the coset of 3370 they follow them only approximately, with the line $\{4, 5, 6, 8, 10\}$ and every entry within $7.6 \cdot 10^{-3}$. The cosets of 2139 and 3370 are even and contain no line. The lines used to weight them lie in the odd cosets of 1993 and 696, and adding the line to a word of either even coset gives a word in the coset of 3986, at distance 5. So the reduced weights weaken exactly the conflicts with the coset of 3986. On the other three cosets the weights are irregular, and each representative's entries lie between 0.82 and 1.15 times their root mean square. The weights also vary slightly between the 32 orbits of Σ_0 inside a coset.

We do not have a closed form for the irregular weights. They were found by numerical optimization of the conditions of the previous subsection, with a target cosine slightly below $1/2$, and then rounded to integers at scale about $2 \cdot 10^8$. The proof does not depend on how they were found. Replacing the weights on these three cosets by the exact line weights would not shorten the proof, because the irregular cosets would still need a computer check.

The total is $27707 + 3072 = 30779$.

4.4. Proof.

Lemma 4.2. *Every S_h , $h \in \Sigma$, maps B' onto itself.*

Proof. Every word of Σ lies in D , so it meets every octad of $\mathcal{O}$ in an even number of coordinates. A sign change S_h maps roots to roots. On an octad vector with support o it changes an even number of signs, so the number of minus signs on o stays odd. So $S_h(B) = B$. Since S_h fixes the 13 deleted points, $S_h(B') = B'$. $\square$

Each S_h is orthogonal and $S_h S_{h'} = S_{h+h'}$. Hence

$$\langle S_h r_i, S_{h'} r_j \rangle = \langle r_i, S_{h+h'} r_j \rangle, \quad \langle S_h r_i, b \rangle = \langle r_i, S_h b \rangle.$$

With Lemma 4.2, every pair involving a new point is determined by one of these finite lists:

- (i) the $\binom{192}{2} \cdot 16 + 192 \cdot 15 = 296\,256$ products $\langle r_i, S_h r_j \rangle$ with $i \leq j$, excluding $i = j$, $h = 0$;
- (ii) the $192 \cdot 27707 = 5\,319\,744$ products $\langle r_i, b \rangle$ with $b \in B'$.

Proposition 4.3. *Let $L = 2 \cdot 10^8$. For every product $\langle x, y \rangle$ in lists (i) and (ii), $\langle x, y \rangle \leq (\frac{1}{2} - \frac{1}{L})|x||y|$. In particular every such pair satisfies (1).*

Proof. This is a finite computation. For $\langle x, y \rangle > 0$ the inequality is equivalent to $(2L\langle x, y \rangle)^2 \leq (L-2)^2|x|^2|y|^2$. For list (ii), Lemma 2.2 gives the maximum of $\langle r_i, b \rangle$ over the fiber over each octad other than Q and over the roots in closed form, and the 115 retained points on Q are checked one by one. We verified both lists in exact integer arithmetic. (Since $\Sigma = \Sigma_0 \cup (w + \Sigma_0)$, it is equivalent to check the 384 vectors r_i , $S_w r_i$ under Σ_0 ; our program does this.) A second program, written independently and without the reduction, verified (1) for all 473 658 031 pairs, and so does the formal proof of Section 5. $\square$

Base pairs are compatible by Lemma 2.1. So all 30779 points are pairwise compatible, and they span $\mathbb{R}^{21}$ because B' contains all 840 roots. This proves Theorem 1.1 for $n = 21$.

4.5. Remarks. The smallest value of $\frac{1}{4} - \cos^2$ over pairs involving a new point is

$$\frac{105355791297}{799999998594867688} \approx 1.32 \cdot 10^{-8},$$

attained by a new point and an octad vector on an octad other than Q . The tightest pair of two new points has $\frac{1}{4} - \cos^2 \approx 1.98 \cdot 10^{-8}$; their sign vectors are at Hamming distance 5. All 12 975 933 contacts are between base points.

The deletion is necessary for these representatives. For each of the 13 deleted points, the largest cosine with a new point lies between 0.50444 and 0.50496, and each of the other 115 odd patterns on Q has cosine at most 0.49999998 with every new point. The group $\{S_h\}$ fixes every vector supported on Q , so different orbits can have different entries on the coordinates of Q without breaking the symmetry.

The role of w . The following observations explain the choice of w ; the proof above does not use them.

- (a) For every $h \in D$ that vanishes on Q , the proof of Lemma 4.2 shows that S_h maps B' onto itself and fixes each of the 13 deleted points.
- (b) So if a set A of new points is compatible with B' and with itself, then so is $S_h(A)$, and $A \cup S_h(A)$ needs no further deletions. The only new conditions are the pairs between A and $S_h(A)$.
- (c) Exactly 32 words of D vanish on Q , and 8 of them form Σ_0 . So up to Σ_0 there are three choices of h besides Σ_0 itself. For $A = \{S_h r_i\}$, $h \in \Sigma_0$, the largest cosine between A and $S_h(A)$ is 0.4746 on the class of w , but 0.5117 and 0.9794 on the other two classes, which therefore fail.
- (d) The vectors $r_1, \dots, r_{192}$ are half of a table of 384 representatives under Σ_0 in an earlier, unpublished 30763-point configuration found by an AI agent, which deleted 29 points on Q . Its 384 orbits fall into two halves of 192 with no pair across them of cosine above 0.491, and the 29 deleted patterns split 13 and 16 between the halves. Keeping the half A needs only the 13 deletions, and (b) with $h = w$ supplies the other 1536 points.
- (e) S_w moves each coset of C to one of opposite parity, which is why all eight cosets appear (Section 4.3). By Lemma 4.1, flat sign vectors from cosets of opposite parity conflict at cosine $11/21$; with these weights the largest cross cosine is 0.4746.

Items (a) and (b) follow from the proof of Lemma 4.2; the numbers in (c)–(e) were computed numerically.

With the base held fixed, each new point may be moved independently to any unit vector within chord distance $1/(8 \cdot 10^8)$ of its position, and the result is again a kissing configuration, because each inner product then changes by at most $2/(8 \cdot 10^8) < 1/L$. So with B' fixed,

the kissing configurations near this one contain an open subset of $(S^{20})^{3072}$, of dimension $3072 \cdot 20 = 61440$. We do not know whether this freedom allows a 30780th point.

5. FORMAL VERIFICATION

We formalized Theorem 1.1 in Lean 4 [10] with Mathlib [9], using Lean 4.33.1 and Mathlib at tag `v4.33.1`. For $(n, N) = (19, 12276)$ and $(21, 30779)$ the main theorems state that there is a finite set S in `EuclideanSpace ℝ (Fin n)` with

$$|S| = N, \quad \|x\| = 1 \quad (x \in S), \quad \langle x, y \rangle \leq \frac{1}{2} \quad (x \neq y \in S).$$

These are the existence statements that the bounds $\tau_n \geq N$ mean; we do not define τ_n in Lean.

The development has two layers. A computational layer builds each configuration as an array of integer rays from its compact description: in dimension 19 the generators of E , the base of Section 3.5, the generators of Γ and the 14 representatives; in dimension 21 the Golay polynomial, the 13 deleted patterns, the generators of Σ and the 192 representatives. It defines a Boolean test of (1) on every pair and a test that every ray has n coordinates and is nonzero. A verified layer proves, with kernel-checked proofs, that if both tests return true then the normalized rays are N distinct unit vectors with pairwise inner products at most $1/2$.

The tests are evaluated with `native_decide`, which compiles the Boolean function, runs it, and if the result is true adds an axiom asserting that it evaluates to true. Each main theorem depends on `propext`, `Classical.choice`, `Quot.sound` and three such axioms, for the size of the array, the nonzero rays and the pairs. Trusting them means trusting that Lean’s compiler and its arbitrary-precision integer runtime compute what the definitions say; a kernel evaluation of 473 million pairs is out of reach.

Only the conclusion is formally verified. Lean does not verify that the arrays it builds match the descriptions in Sections 3 and 4, and it does not verify our case analysis. We checked separately, by computer, that the rays built in Lean and the coordinates in the accompanying data define the same 12276 and 30779 directions.

6. CONCLUSION

We have shown that $\tau_{19} \geq 12276$ and $\tau_{21} \geq 30779$, improving the best published lower bounds of 11948 and 29768. We leave two questions about the 19-dimensional construction open. Can a second orbit of eight deleted points be restored, under Γ or under a smaller group with a larger table? And does the table of Section 3.5 have a closed form? Remark 3.9 suggests one: outside the orbit of r_6 it is close to the additions of Section 3 with new tail factors. In 21 dimensions two questions remain. Can some of the 13 deleted points be restored by moving the 192 representatives? And does the step of Section 4.5, keeping one of two decoupled halves of a table and refilling the other with a sign change that fixes the deleted octad, work in 20 or 22 dimensions?

USE OF ARTIFICIAL INTELLIGENCE

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions, write the Lean formalization, and draft this paper.

DATA AVAILABILITY

The coordinates of both configurations and of the 12268-point configuration of Proposition 3.4, the representatives of Sections 3.5 and 4, short plain-text descriptions of both constructions, the verification programs and the Lean project are available at

<https://github.com/Felpix-Studios/D19-D21-Kissing>

There the 21-dimensional representatives are stored for use under Σ_0 , as the 384 vectors r_i and $S_w r_i$; this gives the same set of points.

APPENDIX A. REPRESENTATIVES

Table 5 lists the 14 representatives of Section 3.5, in the order of the accompanying data. The axis representatives $r_1, \dots, r_6$ have tails along $-f_1, -f_2, +f_3, -f_3, +f_2$ and $+f_1$; r_6 is the representative of Remark 3.9. Each is written as two rows, the first with head coordinates 2 to 14 and the second with head coordinates 15 to 20 and the tail.

TABLE 5. The 14 representatives r_i in dimension 19. Columns 2, $\dots$, 20 are the head coordinates, and f_1, f_2, f_3 the tail.

	2	3	5	6	8	10	11	12	13	14
	15	16	17	18	19	20	f_1	f_2	f_3	
r_1 , axis	1638	1638	-1638	-1638	-1638	1638	1638	-1638	1638	1638
	-1638	1638	-1638	1638	1638	1638	-4588	0	0	
r_2 , axis	-1633	1633	1633	-1633	-1633	-1633	-1633	1633	1633	1633
	-1633	1633	1633	1633	1633	1633	-102	-4616	1	
r_3 , axis	1633	1633	1633	-1633	-1633	-1633	1633	1635	1633	1633
	1633	1635	-1633	1633	1633	1633	-101	0	4616	
r_4 , axis	-1633	-1633	-1633	-1633	-1633	-1633	-1633	-1633	1633	1633
	1633	1633	-1633	1633	-1633	1633	-100	0	-4616	
r_5 , axis	1633	-1633	-1633	-1633	-1633	-1633	1633	-1633	1633	1633
	-1633	1633	1633	1633	-1633	1633	-102	4616	1	
r_6 , axis	-1777	-1453	1777	-1453	-1453	1777	-1777	1777	1777	1777
	-1777	1777	-1777	1777	-1453	1777	4203	0	0	
r_7 , diagonal	-1881	-1881	1881	-1881	-1881	1881	1881	-1881	1881	1881
	1881	1881	-1881	1881	1881	1881	-1568	-1568	-1568	
r_8 , diagonal	1881	-1881	1881	-1881	-1881	1881	-1881	-1881	1881	1881
	-1881	1881	1881	1881	1881	1881	-1568	1568	1568	
r_9 , diagonal	-1858	-1885	-1885	-1858	-1858	-1856	1885	1856	1885	1858
	-1885	1856	-1856	1885	1885	1885	1593	-1649	1649	
r_{10} , diagonal	1881	-1881	-1881	-1881	-1881	-1881	-1881	1881	1881	1881
	1881	1881	1881	1881	1881	1881	-1568	-1568	1568	
r_{11} , diagonal	-1885	1858	1884	-1885	-1885	-1857	1859	-1857	1884	1885
	1884	1857	1857	1884	-1858	1859	1592	1650	-1650	
r_{12} , diagonal	1881	1881	1881	-1881	-1881	-1881	-1881	-1881	1881	1881
	-1881	1881	-1881	1881	-1881	1881	-1568	1568	-1568	
r_{13} , diagonal	-1859	1859	-1884	-1884	-1884	1884	1885	1857	1884	1859
	-1857	1857	1884	1857	-1858	1885	1592	-1650	-1650	
r_{14} , diagonal	1885	1885	-1885	-1858	-1858	1885	-1858	1856	1885	1885
	1856	1856	-1885	1856	-1885	1858	1593	1649	1649	

Table 6 lists, for dimension 21, the first representative on each of the six cosets met by $r_1, \dots, r_{192}$, numbered as in the accompanying data. Each is written as three rows of seven entries, coordinates 1 to 21.

TABLE 6. One representative r_i per coset of C in dimension 21.

r_1 , coset 696	246234381	219980925	209785187	-202482891	-224617629	-218092629	216543425
	-219980925	217704730	-239855454	239957324	216009092	220121191	204061050
	236622028	201715279	231423123	216136122	224959675	180951525	181953903
r_{33} , coset 1993	235702242	-235702246	235702245	235702249	-235702248	-235702244	-235702246
	-149071276	235702243	235702244	-235702242	-235702243	-235702244	-235702248
	149071272	149071276	235702246	149071274	149071274	235702248	235702248
r_{65} , coset 2787	225351549	182591213	247281647	-228539751	-187676135	235642930	-216731283
	-207740975	-197342048	231905242	225679867	188757907	-222180761	-210430962
	204605411	249283059	215561908	215633807	221449791	227383590	225778340
r_{97} , coset 1393	214926831	-190063656	233096365	-230585945	200371893	241359405	-235160970
	205103483	182738919	-215303842	-210489005	-184471317	-226272166	-204197249
	239084407	234925920	224054811	221607207	221607197	229029592	224055305
r_{129} , coset 2139	149071274	235702247	149071275	235702251	149071276	-235702243	-235702244
	235702247	-235702244	-235702243	235702242	235702244	-235702243	-235702248
	235702242	235702248	149071275	149071274	235702242	235702247	235702249
r_{161} , coset 3370	235368146	-235367792	235367792	-149956790	149584899	-150128199	235368272
	149956667	-235221677	150184393	-235795937	-235221410	235740042	235392962
	235917546	235399007	235796162	235739740	235942608	235368197	235399408

REFERENCES

- [1] H. Cohn, *Kissing numbers*, table, <https://cohn.mit.edu/kissing-numbers/>, accessed 4 October 2026.
- [2] H. Cohn and A. Li, *Improved kissing numbers in seventeen through twenty-one dimensions*, preprint, 2024, version 3, 2026. arXiv:2411.04916v3.
- [3] J. H. Conway and N. J. A. Sloane, *Sphere packings, lattices and groups*, third edition, Grundlehren der mathematischen Wissenschaften 290, Springer, New York, 1999.
- [4] B. S. Ho, *A new lower bound for the kissing number in 19 dimensions*, preprint, 2026. arXiv:2603.10425v2.
- [5] J. Leech, *Notes on sphere packings*, Canadian J. Math. **19** (1967), 251–267.
- [6] J. Leech and N. J. A. Sloane, *Sphere packings and error-correcting codes*, Canadian J. Math. **23** (1971), no. 4, 718–745.
- [7] N. Leijenhorst and D. de Laat, *Solving clustered low-rank semidefinite programs arising from polynomial optimization*, Math. Program. Comput. **16** (2024), no. 3, 503–534.
- [8] F. J. MacWilliams and N. J. A. Sloane, *The theory of error-correcting codes*, North-Holland, Amsterdam, 1977.
- [9] The mathlib Community, *The Lean mathematical library*, in Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.
- [10] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in Automated Deduction, CADE 28, Lecture Notes in Comput. Sci. 12699, Springer, 2021, 625–635.